

Introduction to Cryptography

Outline

2. Introduction to Cryptography.

What Is Cryptography?

Breaking an Encryption Scheme.

Types of Cryptographic Functions.

Secret Key Cryptography.

Public Key Cryptography.

Hash Algorithms.

Cryptography

κρυπτο γραφη

The art of secret writing.

The art of mangling information into apparent unintelligibility
in a manner allowing a secret method of unmangling.



Fundamental Tenet of Cryptography

*If lots of smart people have failed to solve a problem,
then it probably won't be solved (soon).*

Cryptography Definitions

- Messages:

- Plaintext
- Ciphertext

- Ingredients:

- Algorithm(s)
- Key(s)

- Players:

- *Cryptographer*: invents clever algorithms
- *Cryptanalyst*: breaks clever algorithms



Cryptography

- Algorithm
- Key(s) = secret value(s)
- OK for good algorithm to be public
 - Not OK to use bad algorithm
 - “Sunlight is the best disinfectant”
 - Algorithm without key does not help unmangle the information

Computational Issues

- Algorithm should be reasonably efficient
- Security depends on how hard it is to break
- Combination lock
 - 3 number sequence (2R, 1L, 0R), #s between 1-40
 - Possible combinations:



Computational Issues

- Algorithm should be reasonably efficient
- Security depends on how hard it is to break
- Combination lock
 - 3 number sequence (2R, 1L, 0R), #s between 1-40
 - Possible combinations: $40^3 = 64,000$
 - 10 seconds per sequence: 178 hours (/ 2 = 89)



Computational Issues

- Algorithm should be reasonably efficient
- Security depends on how hard it is to break
- Combination lock
 - 3 number sequence (2R, 1L, 0R), #s between 1-40
 - Possible combinations: $40^3 = 64,000$
 - 10 seconds per sequence: 178 hours ($/ 2 = 89$)



Computational Issues

- Algorithm should be reasonably efficient
- Security depends on how hard it is to break
- Combination lock
 - 3 number sequence (2R, 1L, 0R), #s between 1-40
 - Possible combinations: $40^3 = 64,000$
 - 10 seconds per sequence: 178 hours (/ 2 = 89)
 - 4 number sequence, 13 seconds per sequence



Computational Issues

- Algorithm should be reasonably efficient
- Security depends on how hard it is to break
- Combination lock
 - 3 number sequence (2R, 1L, 0R), #s between 1-40
 - Possible combinations: $40^3 = 64,000$
 - 10 seconds per sequence: 178 hours (/ 2 = 89)
 - 4 number sequence, 13 seconds per sequence
 - $40^4 = 2,560,000$
 - 9,244 hours



Computational Issues

- Algorithm should be reasonably efficient
- Security depends on how hard it is to break
- Combination lock
 - 3 number sequence (2R, 1L, 0R), #s between 1-40
 - Possible combinations: $40^3 = 64,000$
 - 10 seconds per sequence: 178 hours (/ 2 = 89)
 - 4 number sequence, 13 seconds per sequence
 - $40^4 = 2,560,000$
 - 9,244 hours



wikiHow
to do anything

Home » Categories » Home and Garden » Home Improvements and Repairs » Tools » Locks and Keys

How to Crack a "Master Lock" Combination Lock

125 authors | 664 revisions | Last updated: November 21, 2012

Publication Issues

- Public or secret algorithms?

Publication Issues

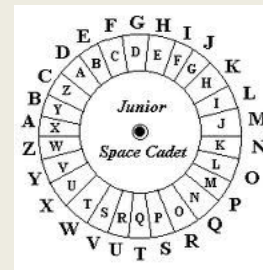
- Public or secret algorithms:
 - OK for good algorithm to be public
 - “Sunlight is the best disinfectant”
 - Reverse engineering, leaks
 - “Free consulting” by cryptanalysts
- Generally:
 - Commercial: public
 - Military: secret

Secret Codes

- Substitution Ciphers
 - Caesar Cipher
 - Vigenere Cipher
 - Simple? Substitution Cipher

Secret Codes

- Caesar cipher
 - Substitute letter 3 letters further on
 - DOZEN ☐ GRCHQ
- Captain Midnight Secret Decoder Rings
 - Substitute letter n letters further on ($n = 1..25$)
 - HAL ☐ IBM ($n = 1$)



Secret Codes

- Caesar cipher
 - Substitute letter 3 letters further on
 - DOZEN ☐ GRCHQ
- Captain Midnight Secret Decoder Rings
 - Substitute letter n letters further on ($n = 1..25$)
 - HAL ☐ IBM ($n = 1$)
- Monoalphabetic cipher
 - Arbitrary mappings ($26! = 4.03291461 \times 10^{26}$)
 - 1 ms / try ☐ 10M years ... but: letter frequencies

Earth Day Cryptograms
by HolidayPartyDecorations.com

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
23																									

E	A																								
14	23	7	16	12	15	23	13	4	19	4	22	16	14	22	15	14	15	16	18	8	23	25	14		
	A	A	E	O																					
6	19	23	9	23	7	14	18	1	12	18	9	9	14	4	22	1	2	6	14	22	5	14			
	E	O		E																					
7	14	19	18	6	7	5	14	19	1	7	18	8	18	6	7	20	2	23	22	14	16				

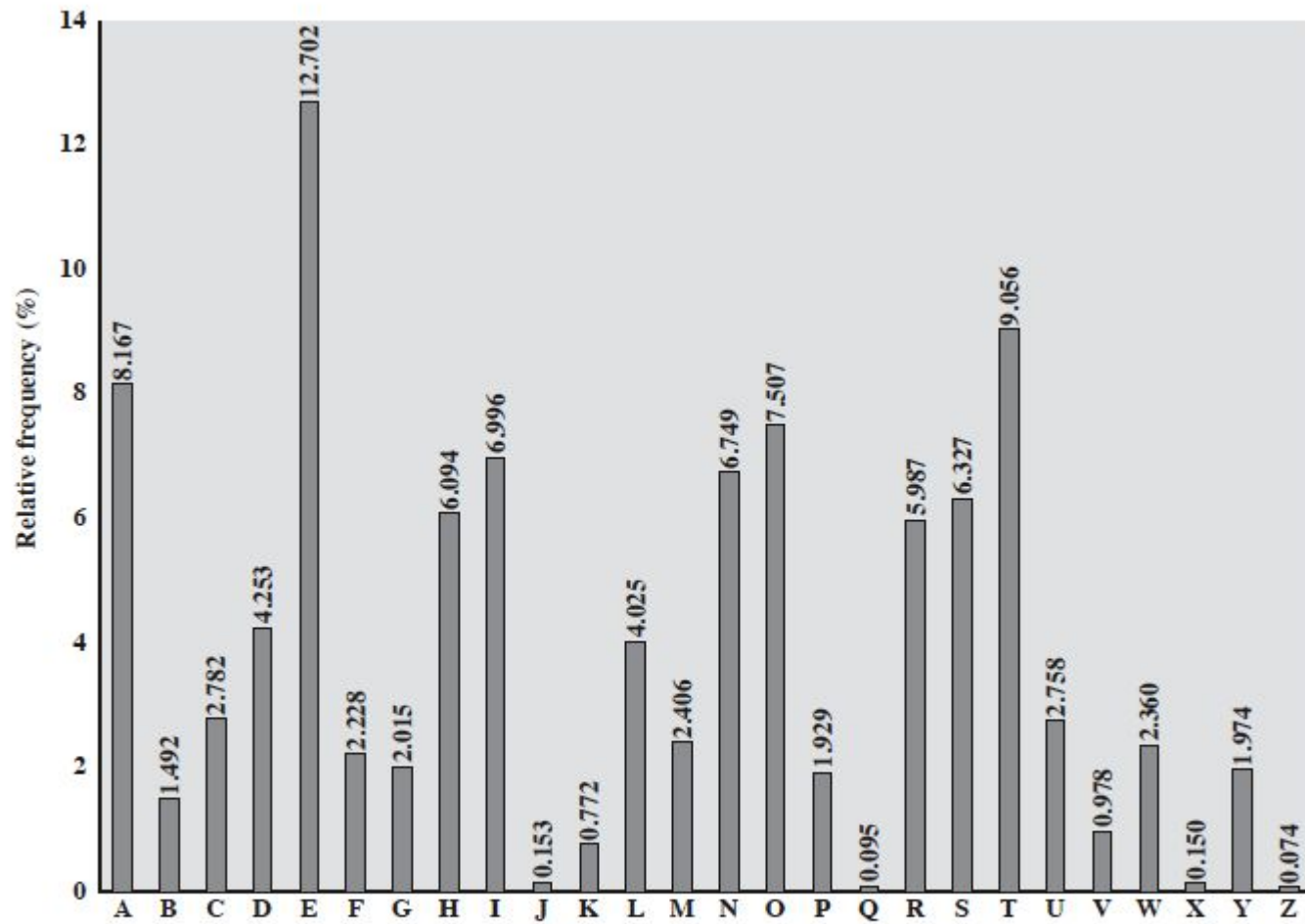


Figure 2.5 Relative Frequency of Letters in English Text

[from *Stallings, Cryptography & Network Security*]

Secret Codes

- Polyalphabetic cipher
 - Where the same character can be mapped onto different characters
 - E.g. Vigenere Cipher
 - Can Frequency Analysis work here?

- Vigenere Cipher Encryption
 - Plaintext: “GEEKS” Key: “FIRST”
 - Ciphertext: ?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

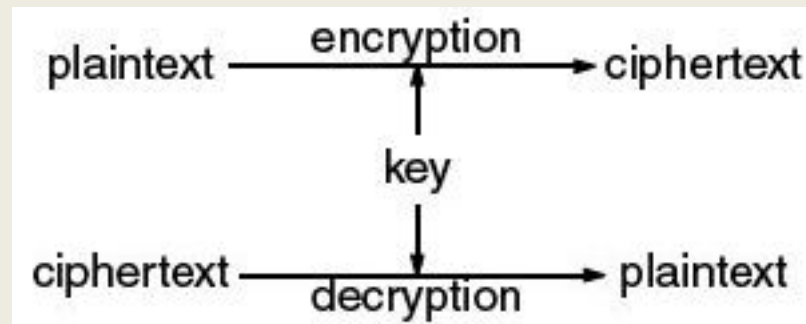
Breaking an Encryption Scheme

- Ciphertext only
 - Try all possible keys, look for intelligibility
 - Need sufficiently long ciphertext
 - XYZ = ? The hot cat was sad but you may now sit and use her big red pen.
- Known plaintext
 - (plaintext, ciphertext) pair(s)
- Chosen plaintext
 - Have plaintext encrypted, compare expected values

Types of Cryptography

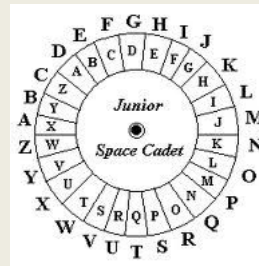
- Public Key
 - Two keys: public & private
- Symmetric Key (aka “Secret Key”)
 - One key: secret (but possibly shared)
- Hash Functions
 - No keys

Symmetric Key Cryptography



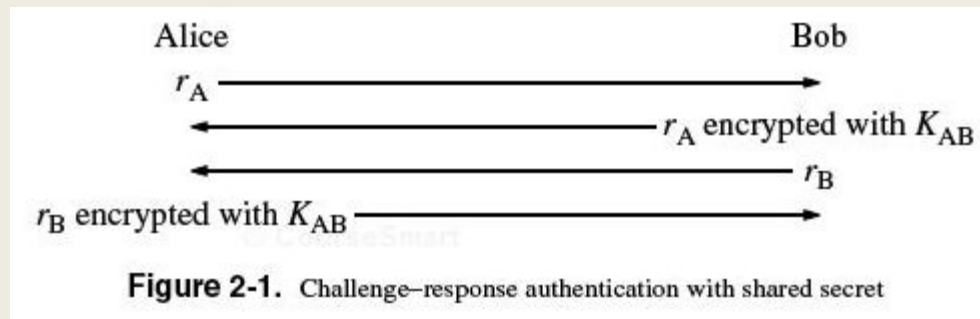
AKA secret key cryptography

AKA conventional cryptography



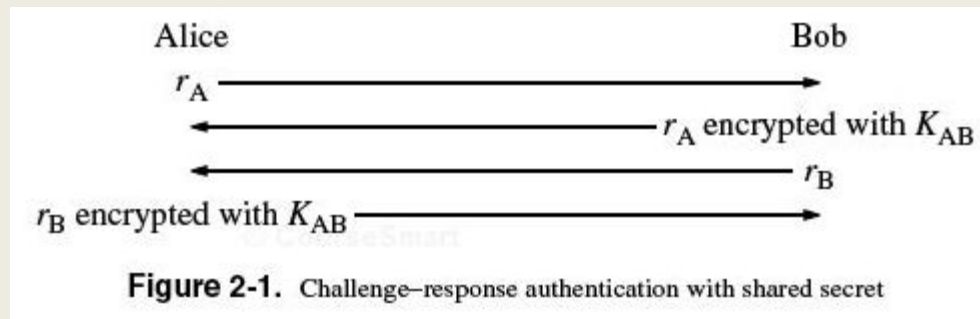
Symmetric Key Applications

- Transmission over insecure channel
 - Shared secret (transmitter, receiver)
- Secure storage on insecure media
- Authentication
 - *Strong* authentication: prove knowledge without revealing key



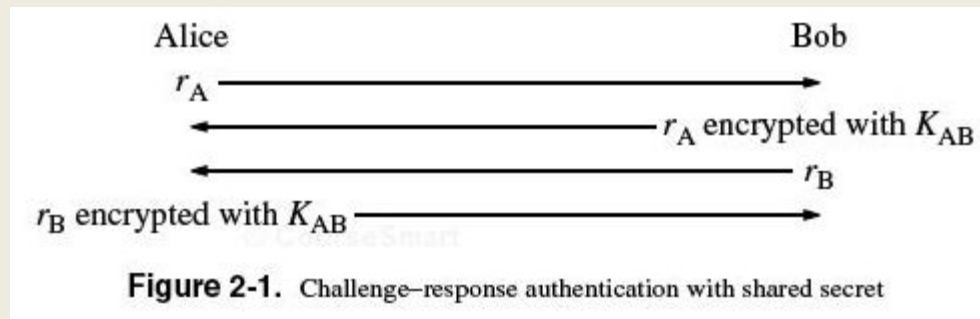
A simple example

- $K_{AB} = +3$ (Caesar cipher), known by Alice & Bob
- $r_A = \text{"marco"}$
 - r_A encrypted with K_{AB} :
- $r_B = \text{"polo"}$
 - r_B encrypted with K_{AB} :



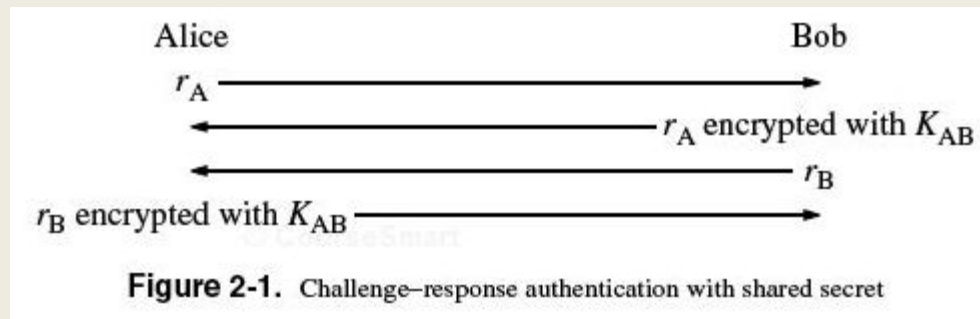
A simple example

- $K_{AB} = +3$ (Caesar cipher), known by Alice & Bob
- $r_A = \text{"marco"}$
 - r_A encrypted with K_{AB} : "pdufr"
- $r_B = \text{"polo"}$
 - r_B encrypted with K_{AB} : "srord"

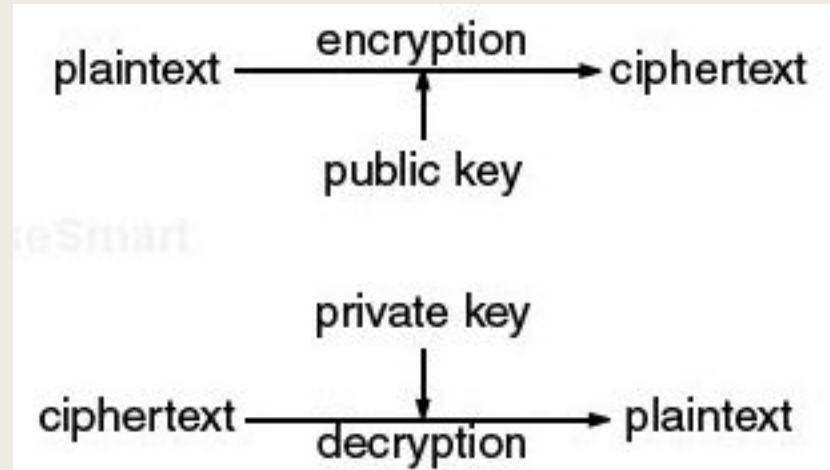


A simple example

- $K_{AB} = +3$ (Caesar cipher), known by Alice & Bob
- $r_A = \text{"marco"}$
 - r_A encrypted with K_{AB} : "pdufr"
- $r_B = \text{"polo"}$
 - r_B encrypted with K_{AB} : "srer"
- ("marco", "pdufr"), ("polo", "srer")



Public Key Cryptography



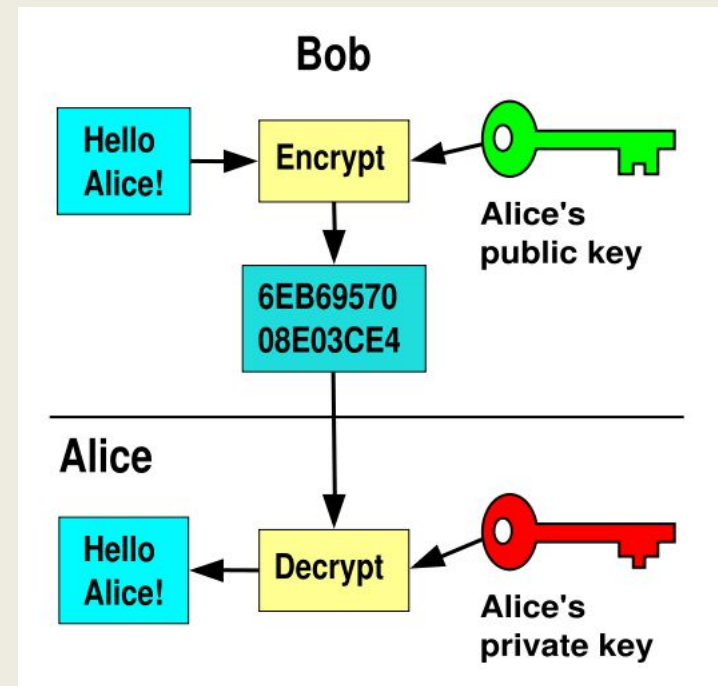
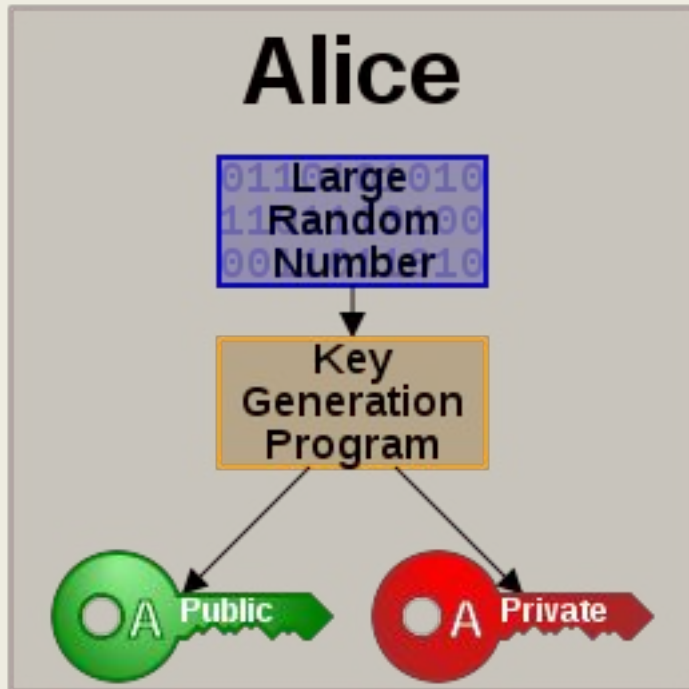
AKA asymmetric cryptography

AKA unconventional cryptography (?)

Public key: published, ideally known widely

Private key (NOT “secret key”): not published

Public Key Cryptography

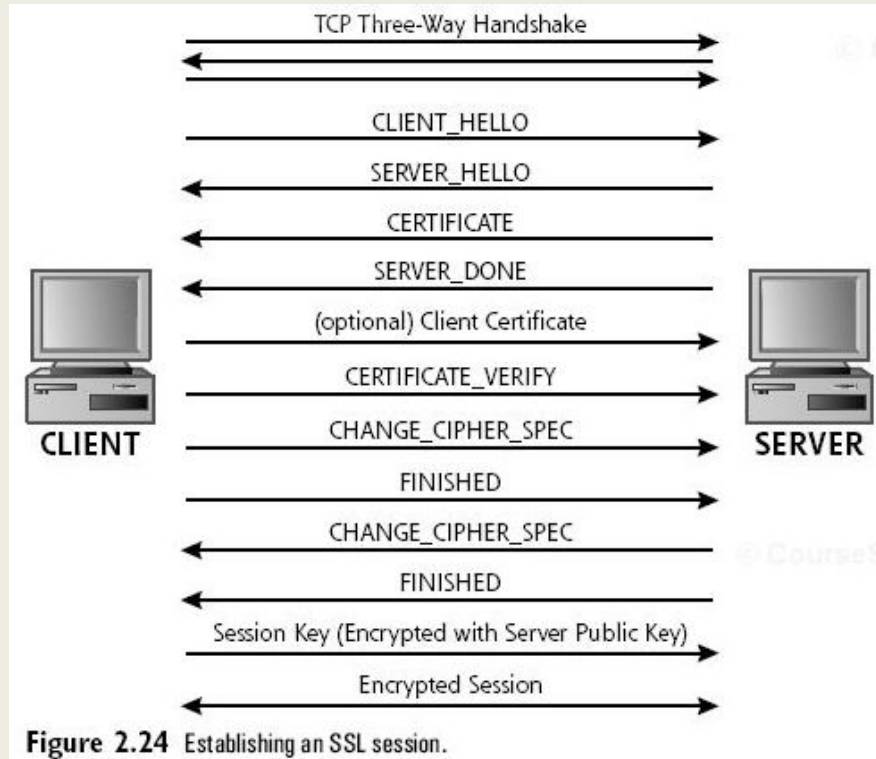


http://en.wikipedia.org/wiki/Public_key_cryptograph

Public Key Cryptography Issues

- Efficiency
 - Public key cryptographic algorithms are orders of magnitude slower than symmetric key algorithms
- Hybrid model
 - Public key used to establish temporary shared key
 - Symmetric key used for remainder of communication

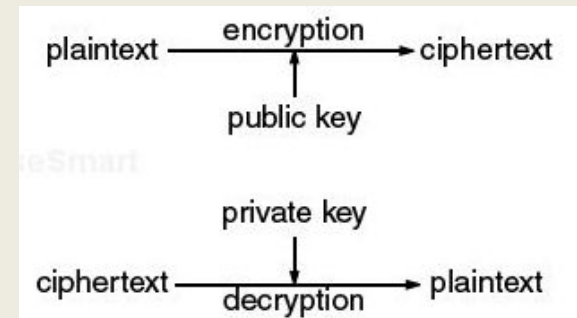
SSL / TLS



- **CLIENT_HELLO:**
 - Available crypto & compression algorithms
 - Highest SSL/TLS protocol version
 - SSL Session ID
 - Random data
- **SERVER_HELLO**
 - Specific crypto & compression
 - Specific SSL version
 - SSL Session ID
 - Random data
- **CERTIFICATE**
 - Server's public encryption key
- **Session Key**
 - Server's public encryption key + random data

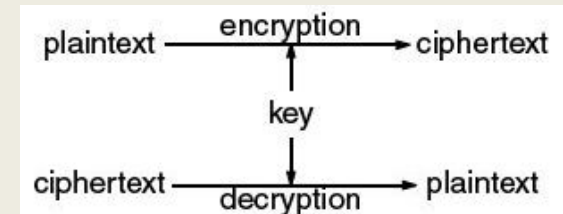
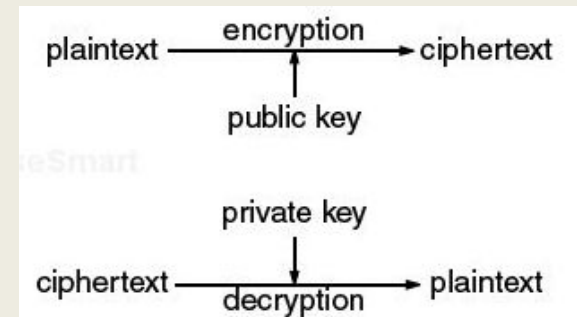
Secure Storage on Insecure Media

- Option 1:
 - Encrypt with public key
 - Decrypt with private key



Secure Storage on Insecure Media

- Option 1:
 - Encrypt with public key
 - Decrypt with private key
- Option 2:
 - Generate random secret key
 - Encrypt with that secret key
 - Encrypt secret key with public key

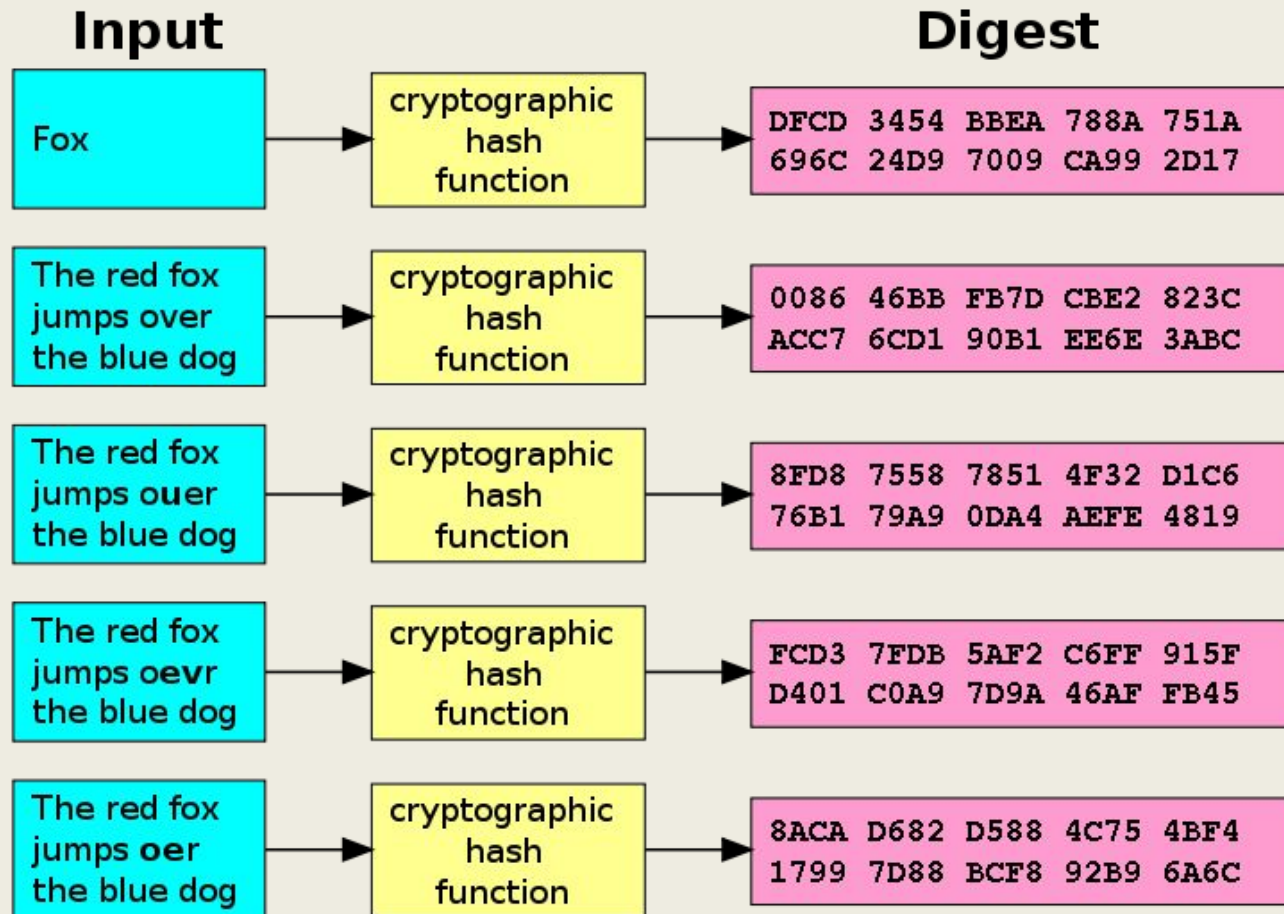


Hash Algorithms

- Message digests / one-way transformations
 - easy to compute a hash value for any given message
 - infeasible to find a message that has a given hash
 - infeasible to modify a message without hash being changed
 - infeasible to find two different messages with the same hash

http://en.wikipedia.org/wiki/Cryptographic_hash_function

Hash Algorithms



http://en.wikipedia.org/wiki/Cryptographic_hash_function

An example

This example uses the common unix utility "**md5sum**", which hashes the data on stdin to a 128 bit hash, displayed as 32 hex digits.

Assume the password is "**mysecretpass**" and both the client and the server know this.

The client connects to the server.

The server makes up some random data, say "**sldkfjds**l**fkjweifj**".

The server sends this data to client.

The client concatenates the random data with the password, resulting in "**sldkfjds**l**fkjweifj****mysecretpass**"

The client computes the MD5 hash of this value:

```
$ echo sldkfjdslfkjweifjmysecretpass | md5sum  
4fab7ebffd7ef35d88494edb647bac37
```

The client sends "4fab7ebffd7ef35d88494edb647bac37" to the server.

[The server confirms that it gets the same value when it runs echo "**sldkfjds**l**fkjweifj****mysecretpass**" | md5sum]

<http://www.hcsw.org/reading/chalresp.txt>

md5sum on Linux

```
[joemcc@uw1-320-17 ~]$ cat > test1.txt
```

Testing 1, 2, 3

```
[joemcc@uw1-320-17 ~]$ md5sum test1.txt
```

```
6a8c5c1973dd8ed2df1260297964cd64 test1.txt
```

```
[joemcc@uw1-320-17 ~]$ cp test1.txt test2.txt
```

```
[joemcc@uw1-320-17 ~]$ md5sum test2.txt
```

```
6a8c5c1973dd8ed2df1260297964cd64 test2.txt
```

```
[joemcc@uw1-320-17 ~]$ cat > test3.txt
```

Testing 1, 2, 4

```
[joemcc@uw1-320-17 ~]$ md5sum test3.txt
```

```
5e361a608a1f63b154f259dba0f452dc test3.txt
```

```
[joemcc@uw1-320-17 ~]$ echo "Testing 1, 2, 3" | md5sum
```

```
6a8c5c1973dd8ed2df1260297964cd64 -
```

```
[joemcc@uw1-320-17 ~]$
```

Cryptographic Hash Lifecycle

Life cycles of popular cryptographic hashes (the "Breakout" chart)																							
Function	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Snefru	Green	Green	Green	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red
MD4	Green	Orange	Orange	Orange	Orange	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red	Red
MD5			Green	Green	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Red	Red	Red	Red	Red	Red	Red	Red	Red
MD2			Green	Green	Green	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Red	Red	Red	Red	Red	Red	Red	Red	Red
RIPEMD			Green	Green	Green	Green	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Red	Red	Red	Red	Red	Red	Red	Red	Red
HAVAL-128			Green	Green	Green	Green	Green	Green	Green	Green	Orange	Orange	Orange	Orange	Red	Red	Red	Red	Red	Red	Red	Red	Red
SHA-0				Green	Green	Green	Green	Green	Orange	Orange	Orange	Orange	Orange	Orange	Red	Red	Red	Red	Red	Red	Red	Red	Red
SHA-1						Green	Green	Green	Green	Green	Green	Green	Green	Green	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange	Orange
RIPEMD-128 [1]							Green	Green	Green	Green	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey	Grey
RIPEMD-160							Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Green	Grey	Grey	Grey
SHA-2 family											Green	Green	Green	Green	Green	Green	Green	[2]	Orange	Orange	Orange	Orange	Orange
SHA-3 (Keccak)																				Green	Green	Green	Green

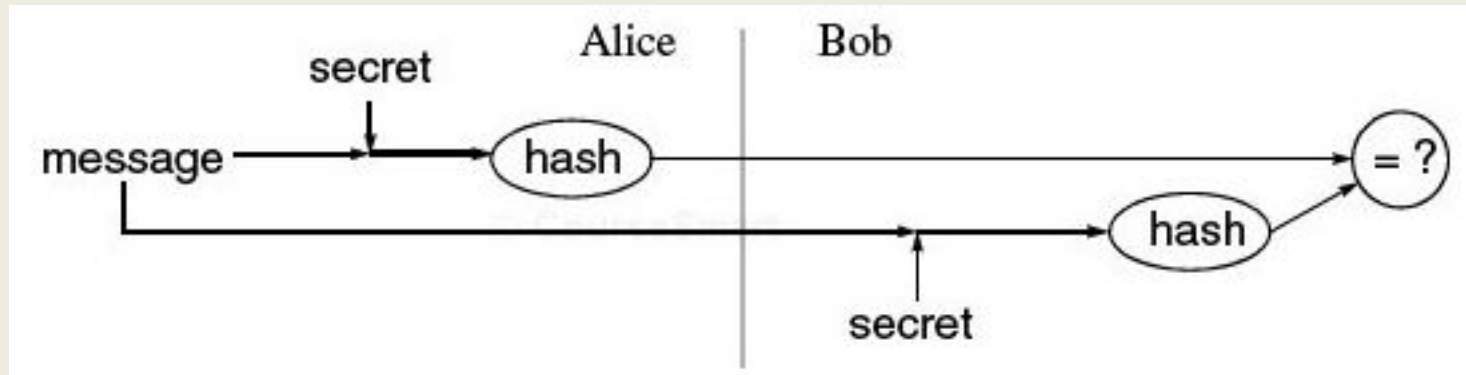
<http://valerieaurora.org/hash.html>

[via

http://www.schneier.com/blog/archives/2011/06/the_life_cycle.html]

Message Integrity

- Keyed hash, shared secret



Password Hashing

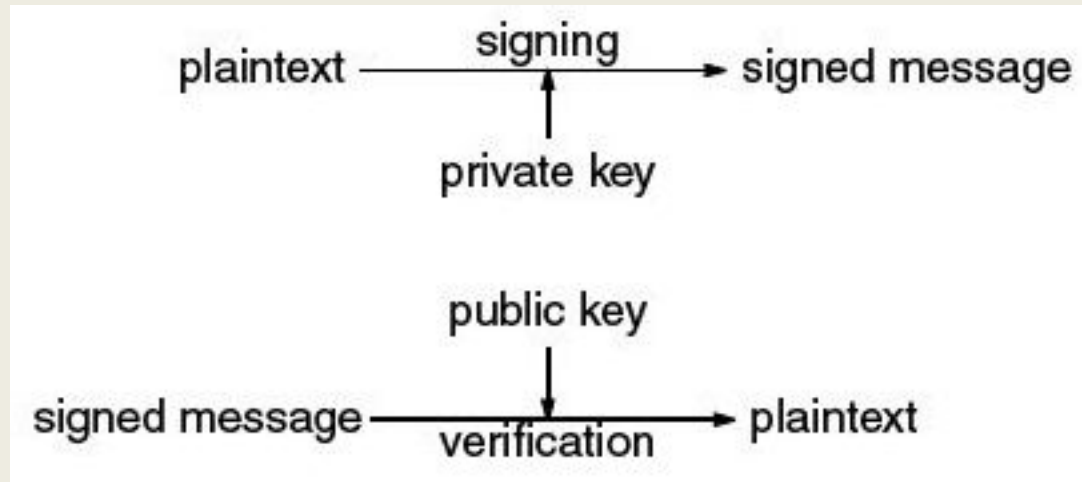
- Only hashes (not passwords) are stored
- Passwords can still be guessed (dictionary)



<http://nakedsecurity.sophos.com/2012/06/06/linkedin-confirms-hack-over-60-of-stolen-passwords-already-cracked/>

fuck	zzzzz	zzzz	zzz	xxxxx	xxxx
xxx	qqqqq	qqqq	qqq	aaaaa	aaaa
aaa	sql	file	web	foo	job
house	work	intranet	controller	killer	game
private	market	coffee	cookie	forever	freedom
student	account	academia	files	windows	monitor
unknown	anything	letitbe	letmein	domain	access
money	campus	explorer	exchange	customer	cluster
nobody	codeword	codename	changea	desktop	security
secure	public	system	shadow	office	supervisor
superuser	share	super	secret	server	computer
owner	backup	database	lotus	oracle	business
manager	temporary	ibmuserpass	nothing	nopassword	nopass
Internet	internet	example	sample	love123	boss123
work123	home123	mypc123	temp123	test123	qwel23
abc123	pu123	root123	pass123	pass12	pass1
admin123	admin12	admin1	password123	password12	password1
default	foobar	foofoo	temptemp	temp	testtest
test	rootroot	root	adminadmin	nypassword	nypass
pass	login	login	Password	password	passwd
nicvbn	szcvb	szcvbz	ncvz	qawzredc	qawex
qlwCe3	qweazdrx	azdrgh	azdrzc	azdrz	azdrz
qweasd	qwerty	qweeq	qweq	ninda	administrator
Admin	admin	alb2c3	lq2w3e	1234qwer	1234abcd
123asd	123qwe	123abc	123321	12321	123123
1234567890	123456789	12345678	1234567	123456	12345
1234	123				
99999999	88888888	77777777	66666666		
9999999	8888888	7777777	6666666		
999999	88888	77777	66666		
9999	8888	7777	6666		
999	888	777	666		
99	88	77	66		
9	8	7	6		
55555555	44444444	33333333	22222222		
5555555	4444444	3333333	2222222		
555555	444444	333333	222222		
5555	44444	33333	22222		
555	4444	3333	2222		
55	444	333	222		
5	44	33	22		
11111111	00000000	987654321			
1111111	0000000	987654321			
111111	00000	87654321			
11111	0000	7654321			
1111	000	654321			
111	00	54321			
11		4321			
1		321			
		21			
		12			

Digital Signatures



Asymmetry:

Signature can only be *generated* by owner/knower of private key

Signature can be *verified* by anyone via public key

Non-repudiation:

Sender cannot prove message (signature) was not sent

Key may have been stolen

